University of Minnesota

Telerobotic Endovascular Skills Assessment (TESA)

Open-H Endoluminal Data Proposal

1. Executive Summary

Endovascular procedures, such as mechanical thrombectomy, depend heavily on the precise navigation of flexible guidewires and catheters through intricate vascular networks. Modern soft-body physics simulators do not capture the extreme edge cases or complex interaction dynamics of these tools, such as those arising under severe torque or sustained mechanical stress that causes cumulative changes to guidewire properties (bending, buckling, and torsion) over many episodes. We propose to address this limitation through scalable real-world data collection across a series of environments or "gyms" that ensure guidewires/catheters perform tasks across a spectrum of complexity, ranging from simple to nearly impossible target poses that require complex nonlinear dynamics like snapping to achieve. To support this goal and bridge the simulation-to-reality gap, we propose to collect and supply a standardized endoluminal dataset (TESA-Data) and low-cost open-source hardware platform (TESA-Bot) to the Open-H ecosystem. Inspired by the impact that the low-cost Fundamentals of Laparoscopic Surgery (FLS) tasks had on the science of surgical technical skill assessment and robotic surgery, we seek a similar set of standardized tasks and hardware for endovascular skills. This will democratize access to both data and hardware, inviting multiple groups to replicate the setup to both generate more data and, crucially, evaluate novel algorithms derived from Open-H efforts on a standardized, widely accessible physical system.

Our dataset will contribute over 30 hours of synchronized robotic telemetry and vision tracking data (Signal Tiers S1/S3) captured within synthetic anatomical vessel cutouts. To gather diverse, high-volume data efficiently without continuous human labor, data collection will be driven primarily by an autonomous reinforcement learning (RL) agent navigating in real time, with human operators stepping in to cover complex bifurcations that the RL model fails to reach. All trials will be conducted on our custom, low-cost (<$100), open-source 2-DOF guidewire and catheter control robot driven by an accessible control board accepting relative step commands $[\Delta\text{insertion}, \Delta\text{rotation}]$ with a synchronized encoder and overhead camera.

To guarantee data integrity, our pipeline incorporates automated computer vision checks to detect motor slippage, re-establish origin baselines, and filter out guidewire loops and buckling. Because all experiments utilize benchtop phantoms derived from open-source anatomical models, the dataset carries zero HIPAA compliance risk or Patient Health Information (PHI) constraints. In alignment with open-science principles, we commit to supplying all generated datasets, codebases, and hardware blueprints under the OpenMDW-1.1 license.

2. Task and domain selection

2.1. Signal Tier: S1 / S3

2.1.1. **S1:** Continuous joint and actuator states, total insertion depth, motor rotation, and commanded step deltas recorded directly from motor encoders.

2.1.2. **S3:** Real-world desktop phantom environment observed via an overhead USB camera, acting as a simulated 2D fluoroscopy frame for tip pose and spline estimation, and motor slippage correction.

2.2. Domains (Gyms): Data collection is conducted using a set of approximately 10 custom physical 3D-printed phantoms, integrated into a desktop rig that holds a camera and robotic guidewire driver (see Figure A.4). Related cerebrovascular representations and navigation methods are described in prior work [1, 2]. The rig, camera, and robotic driver will all be described in more detail in Section 3. Each gym presents unique vessel geometry, providing new navigation challenges for the guidewire robot and helping us to collect a broader dataset.

2.2.1. Gym Fabrication: The phantoms are 3D-printed on a consumer-grade 3D printer, with a thin, semitransparent filament for the lower layer. This creates a translucent base that allows optical tracking systems to easily monitor the guidewire's position and orientation inside the channels. The walls of the phantoms are printed from brightly colored PETG, which provides visual contrast between vessel walls and the guidewire to provide better tracking. These phantoms are all 2-dimensional projections of 3D anatomy, a representation also used in StrokeNav2D [1], though the experimental setup is designed with 3D phantom capability in mind. We support phantoms up to 240x160 millimeters.

2.2.2. Gym Design and Modeling: Each phantom is custom designed to offer unique geometry for guidewire agent learning and data collection. All of these phantoms are directly inspired by the human vascular tree. Some gyms are abstract representations of bifurcations and aneurysms, while others are direct 2D projections of human vascular models, modeled from the NIH Open3D library (https://3d.nih.gov/discover). The abstract models add smooth walls and clean bifurcations to the dataset, while the anatomical projection models offer data similar to real-world settings.

2.2.3. Gym Task Complexity: The gyms and subtasks within them will be scored (and thus labelled) along a continuum of difficulty for a guidewire/catheter to traverse based on a quantitative assessment of operability. This score includes tortuosity measures already in use in clinical settings but will be expanded to include more nuanced analytical assessments inspired by variants of the Steering Law. We hope to include difficulty assessments alongside the rest of our data and to collect data on phantoms with a range of difficulty levels to ensure high data variability.

2.3. Target Volume: 30 to 50 cumulative hours of synchronized action recordings.

2.4. Data Collection: The vast majority of our data will be generated by a reinforcement learning agent, which is being trained to navigate through real-world vascular gyms in real time. RL allows us to collect large quantities of sufficiently randomized data without requiring constant human oversight. We will use human-operated data in some trials for gyms where the geometry is too difficult to be fully explored by the trained RL agent.

2.5. Sensors and Recorded Data Schema: The primary output for each discrete step action is saved as an individual .mp4 video clip, paired with a structured JSON (.json) metadata file. Metadata can be exported to Parquet for downstream dataset integration. The recorded fields are:

2.5.1. **tip_xy:** current position of the guidewire tip, measured in pixels in the camera frame

2.5.2. **spline_points:** list of xy positions that follow the length of the guidewire to provide an approximate shape and guidewire heading estimation. Measured in pixels in the camera frame.

2.5.3. **total_insertion:** total guidewire insertion, measured from motor encoders. Measured in millimeters.

2.5.4. **total_rotation:** total guidewire rotation, measured from motor encoders. Measured in radians. We do not know the absolute starting rotation during data collection, so this is a relative number.

2.5.5. **action_insertion:** the insertion action taken by the robot, expressed as the distance that the robot attempts to insert the guidewire in this action, recorded in millimeters.

2.5.6. **action_rotation:** the rotational action taken by the robot, expressed as the angle through which the robot attempts to rotate the guidewire in this action, recorded in radians.

2.5.7. **goal_xy:** a goal location the guidewire model is attempting to reach. This is primarily used for our reinforcement learning agent as it collects data and is expressed in pixels within the camera frame.

3. Technical Approach

3.1. Open Hardware and Robot Setup:

To minimize barriers to entry and ensure reproducible research in robot-assisted endovascular interventions, our platform uses an accessible, open-hardware design with a total bill of materials (BOM) cost under $100. The mechanical architecture consists of a custom two-degree-of-freedom (2-DoF) mechanism optimized for fine manipulation of standard medical-grade guidewires, building on prior work on telerobotically driven guidewires [3].

3.1.1. **Hardware cost:** < $100 total BOM

3.1.2. **Actuation:** One servo motor controls the linear insertion of the guidewire, translating the guidewire in and out of the gym, along the long axis of the insertion channel. A second servo motor controls the axial rotation of the guidewire.

3.1.3. **Control Interface:** A custom PCB provides a control interface that accepts action vectors in the form:

$$U_t = [\Delta\text{insertion}(mm), \Delta\text{rotation}(rad)]$$

Embedded closed-loop drivers process high-frequency pulse width modulation (PWM) signals to command motor positions. The PCB communicates with the host computer via USB, streaming real-time motor telemetry back to the controller.

3.1.4. **Open Source Release:** For this Open-H Endoluminal challenge, we will open-source the mechanical designs, PCB schematics and control drivers along with our submission to lower the barrier to entry for 2-DoF guidewire data collection.

3.2. Environment Setup:

To ensure high repeatability across trials and maintain rigid spatial calibration between the camera and the vascular phantoms, the experimental setup utilizes a custom-built test rig, which will be open-sourced alongside the rest of the project.

3.2.1. **Hardware Cost:** < $ 100 total BOM (excluding medical-grade guidewire)

3.2.2. **Overhead Camera:** An integrated overhead arm holds the top-down USB web camera at a fixed height and orthogonal angle directly over the light board, providing an unobstructed, simulated single-plane fluoroscopy view of the active workspace.

3.2.3. **Light Board Surface:** The floor of the rig features a uniform light board, which backlights the translucent bottom surface of the anatomical gyms to create a high-contrast visual background. This simplifies real-time tip tracking and enhances the quality of the resulting video.

3.2.4. **Countersunk Alignment System:** The frame of the rig incorporates a precision countersunk mounting hole, which allows various gym cutouts to be swapped out quickly while guaranteeing exact, repeatable alignment relative to the camera frame and robot origin without manual visual re-calibration for each gym.

3.3. Reinforcement Learning Agent:

We use a model-free reinforcement learning agent to explore our gyms and collect large quantities of data by summarizing the problem as a Markov Decision Process (MDP) formulated as follows:

3.3.1. Action Space: Actions, a, are directly mapped to the robot's hardware interface vector U_t :

$$a = [\Delta s, \Delta\theta]$$

Actions are bounded within safe physical operating limits, which provide a scaled step size limit per control cycle. Δs (change in guidewire insertion, in mm) is bounded such that $-50 < \Delta s < 50$, while $\Delta\theta$ (change in guidewire rotation, in rad) is bounded such that $-\pi < \Delta\theta < \pi$.

3.3.2. State Space: To enable ego-centric navigation and control, the observation space $o_t \in O$ at step t is defined relative to the local reference frame anchored at the current guidewire tip. The state representation consists of four modalities:

- **Local Environment Map** (I_{local})**:** A binary image centered at the tip of the guidewire that captures local vessel geometry, boundary constraints, and obstacles. It is taken from the overhead camera and thresholded to only contain the vessel walls as a binary occupancy grid.
- **Guidewire Deformation State** (P_{spline})**:** A set of 5 spatial points sampled along a fitted spline of the distal end of the guidewire, representing the shape and bending state of the wire relative to the tip.
- **Cumulative Actuation Kinematics** (a_{total})**:** The accumulated proximal insertion distance and axial rotation angle $[s, \theta]$, providing context on total displacement to estimate friction effects.
- **Relative Goal Vector** (d_{goal})**:** The heading direction vector pointing from the current tip position toward the target location.

By casting all spatial coordinates (image, spline points, goal heading) into the tip-centered frame, the observation space remains invariant to absolute global translations, improving generalization across different initializations and workspace maps.

3.3.3. Reward Function: The agent is trained using a composite reward function designed to incentivize the agent to explore quickly and with high sample efficiency. It combines a sparse terminal positive reward when the guidewire tip reaches the target location and time penalties for each step the guidewire fails to reach the goal with a dense geodesic progress reward. The progress reward is calculated as the change in potential along a precomputed flood-fill gradient field originating from the goal location. By using geodesic distance within the vessel lumen rather than Euclidean distance, the gradient provides unambiguous guidance around anatomical bends and vessel branches.

3.3.4. Algorithm Selection: We currently use Soft Actor-Critic (SAC) to learn a policy in the continuous action space of guidewire navigation. During the data collection phase, we aim to understand the limitations of SAC and develop other methods.

3.4. Synchronization:

To simplify the camera synchronization problem, we discretize the continuous action space of our robot guidewire and record each action taken by the robot as a separate video file. For each action taken by the robot:

A. **Trigger Camera Start:** The host control system opens an active recording stream from the overhead camera and initializes video buffering.
B. **Execute Actuator Step:** The custom PCB receives an action command vector (insertion, rotation) and drives the servo motors to the target positions.
C. **Wait for Completion:** The driver polls encoder feedback until motor motion ceases.
D. **Stop Capture and Export Data:** The frame stream closes, compiling the step movement into a compressed .mp4 file alongside its associated JSON metadata file. The metadata can be exported to Parquet.

Discretizing the robot's actions over time ensures the data we collect is synchronized between the camera and robot frames.

An LCD readout providing timestamp information will be visible in the camera view. In the event that the above synchronization scheme is determined to be insufficiently accurate, a USB ELP camera with a strobe/trigger line will be connected to the microcontroller to ensure hardware-level synchronization of low-level motor encoder data with video frame times.

3.5. Quality Assurance:

In tight vessel channels or under high friction, the flexible guidewire may buckle, loop back on itself, or slip:

3.5.1. **Slippage Compensation:** We account for motor slippage over time using the overhead camera as a homing mechanism. After a set number of frames in which motor commands are issued but insufficient wire movement is detected, the system retracts the wire to the start of the gym using visual tracking and re-establishes that location as the new zero origin.

3.5.2. **Topological Loop Filtering:** The overhead camera continuously monitors the 1D topological skeleton of the guidewire. Any frames exhibiting self-intersecting loops or structural buckling are flagged as erroneous and discarded.

3.6. Privacy: The vascular gyms used in our study are inspired by real human anatomy but are abstractly engineered and not directly derived from any single patient's vascular tree. The dataset presents no HIPAA compliance concerns and contains no Patient Health Information (PHI).

4. Data Diversity:

The platform will be able to capture physical edge cases of guidewire dynamics that are not accurately represented in current simulation tools, including buckling, looping, torsion, slippage, and straining. Data retained for release will be subject to the quality-assurance filtering described in Section 3.5. A diverse set of vascular anatomies (gyms) will support recording guidewire behavior across varied physical conditions.

4.1. Guidewire actuation is subject to the interactions it has with its surroundings, and to ensure that we are capturing the guidewire's behavior in sufficiently diverse physical situations, we collect data using more than 10 different gyms (see Figure A.3). Each gym has multiple start and end locations for the guidewire, and each of these paths has a different associated difficulty. The gyms will have variable lumen sizes, bifurcation angles, tortuosities, and completely varying structures. Most of these will be direct 2D projections of human anatomy since they provide real anatomical properties that an autonomously navigating guidewire would have to explore, though some of the phantoms will be made more abstract to demonstrate our automatic vessel operability assessment tool and to provide easier tracks for the reinforcement learning agent to train in.

4.2. Within each of these gyms, we will ensure our data is sufficiently diverse by plotting the location of the guidewire tip over time (see Figure A.2). A sufficiently diverse dataset will result in the guidewire exploring all areas of the gym uniformly, which can be quantified with a heatmap (see Figure A.2). Proximal regions of the gym will be more explored than distal regions, but data will be collected from the gym as spatially uniformly as possible. As our reinforcement learning algorithm trains, it will learn to explore the gym in its entirety, and after it is trained, its objective will be to explore the distal branches of the gym, which will result in a fully explored gym. Underexplored areas of the gym will be explored manually by a human operator to ensure that each gym supplies maximum data diversity.

5. Team Qualifications

Team Member	Role	Expertise
Alex Berg	Student	Biomedical data science analyst and robot navigation.
Adam Imdieke	Student	Computer science PhD student at the University of Minnesota researching tactile sensing, contact-aware control, inverse kinematics, and policy learning; co-author of DMD 2026 studies on telerobotic guidewires and cerebrovascular navigation.
Arianna Reischer	Student	Data collection assembly design and project logistics support.
Neev Zeroni	Student	Gym environment mechanical design and anatomical understanding.
Tarun Sehgal	Student	Base Frame mechanical design and setup assembly
Samuel Fisher	Student	Mechatronics module design and motor control electronics
Gabriel Garso	Student	Gyms
Amit Manicka	Student	Mechanical/Electronic interface and Computer Vision
Karthik Desingh	PI	Robot Perception, Manipulation, Learning and AI
Timothy M. Kowalewski	PI	Surgical Robotics, Surgical Skill Assessment
Daniel F. Keefe	PI	Graphics, Visualization, Fluoro-CT Registration
Andrew Grande, MD	PI	Neurosurgeon, Clinical Guidance

Related publications: Team members have published work on telerobotic guidewires [3], vascular roadmaps [2], and 2D cerebrovascular navigation datasets [1].

6. Project Plan (Gantt Chart)

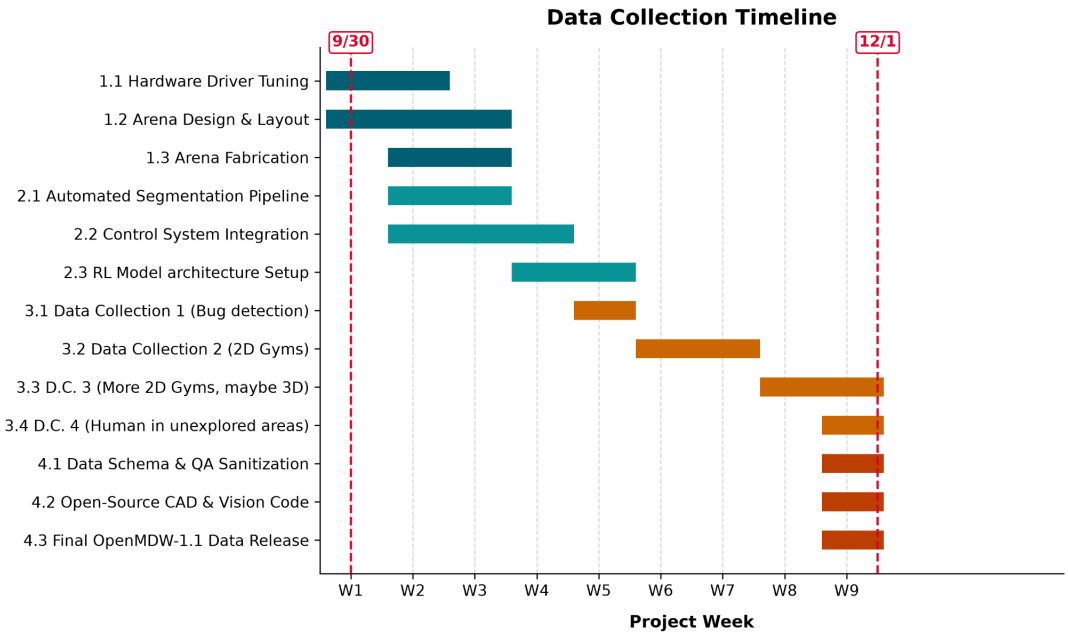


Figure 1: Project timeline.

7. Data rights and IP statement: OpenMDW-1.1

7.1. We explicitly confirm that all software algorithms, firmware codebases, neural network weights, CAD models, electrical schematics, raw video recordings, and associated spatial metadata generated under the Open-H Endoluminal Data Project will be made publicly available without restriction under the OpenMDW-1.1 License (Open Medical Data & Hardware License).

A. Supporting Figures

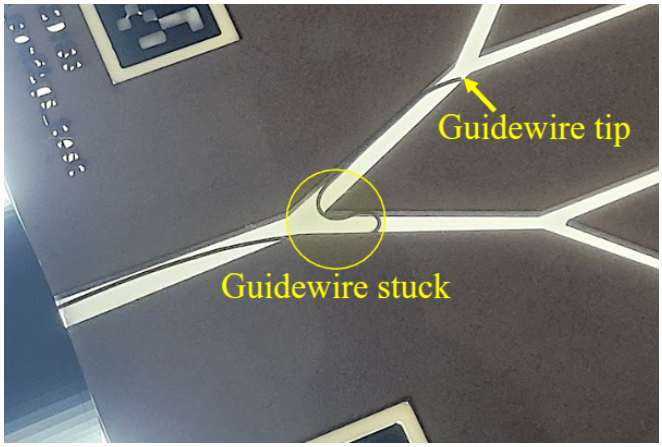


Figure A.1: Guidewire tip and a stuck guidewire configuration.

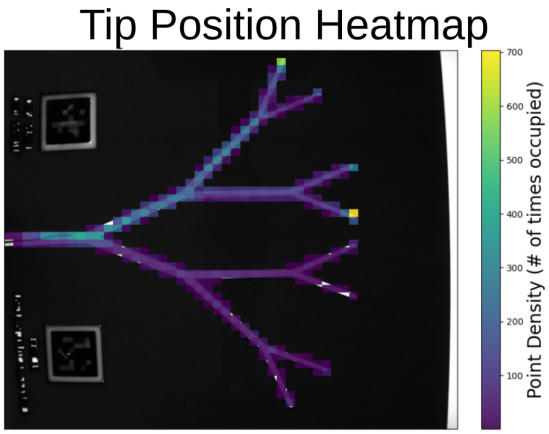


Figure A.2: Tip position heatmap.



Figure A.3: Vascular gym examples.

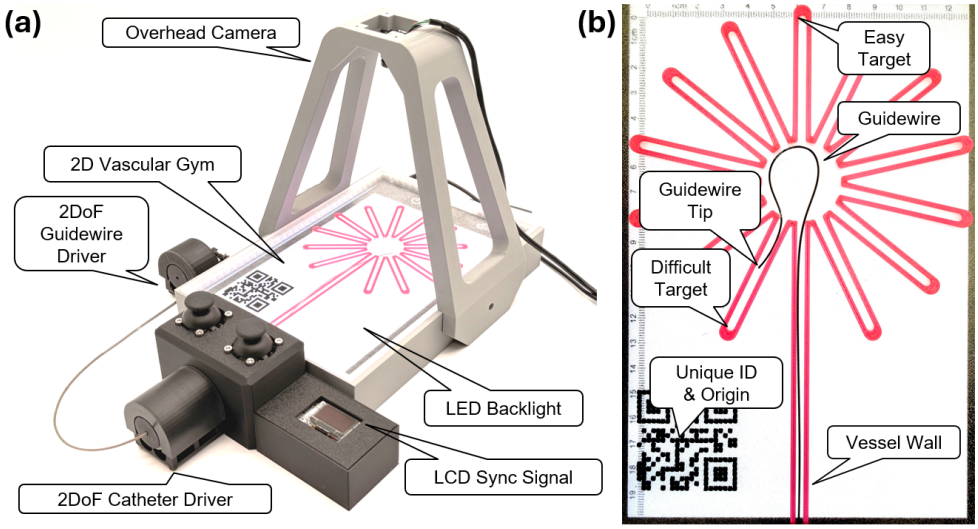


Figure A.4: Experimental setup and overhead camera view.

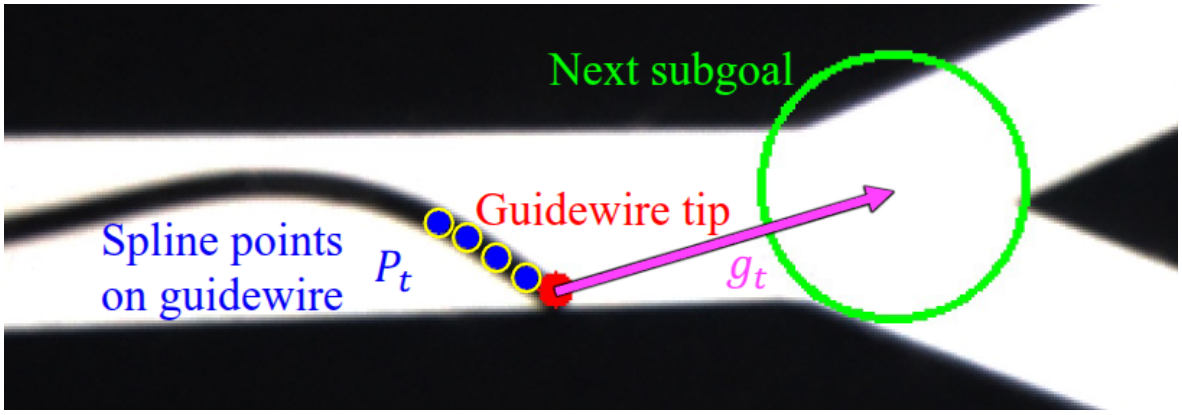


Figure A.5: Spline points, guidewire tip, and next subgoal.

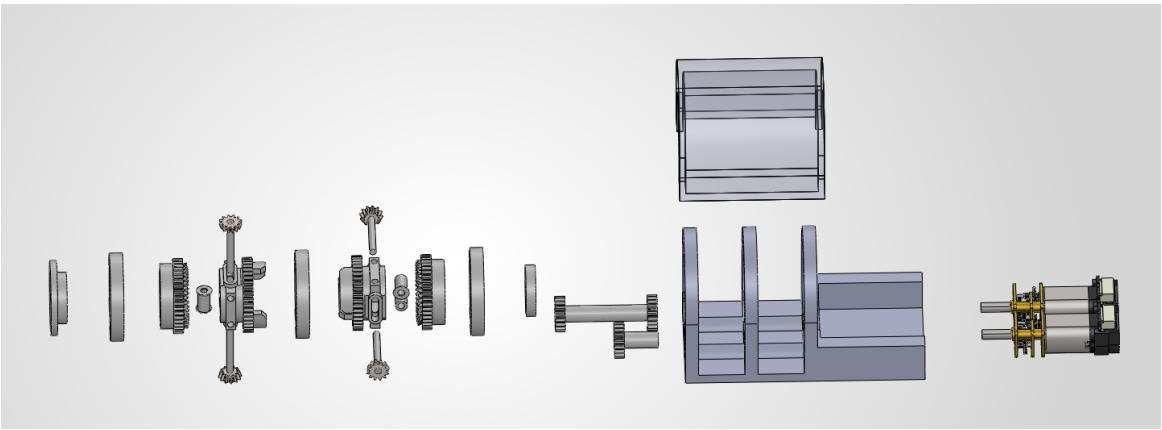


Figure A.6: Exploded view of the guidewire driver.

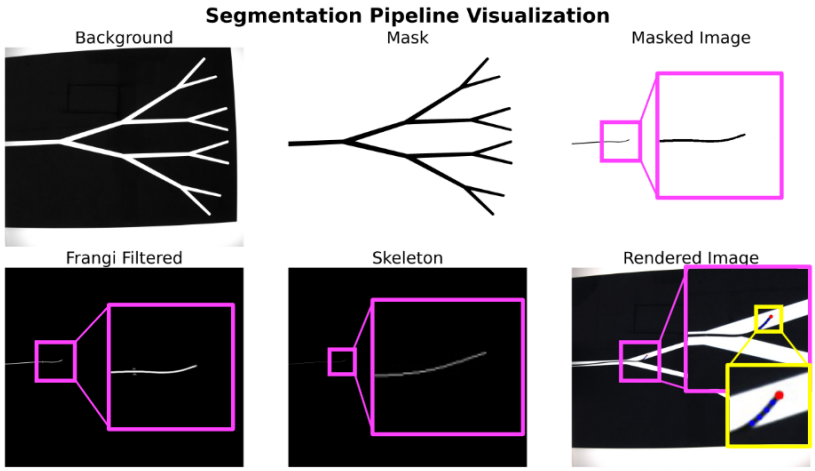
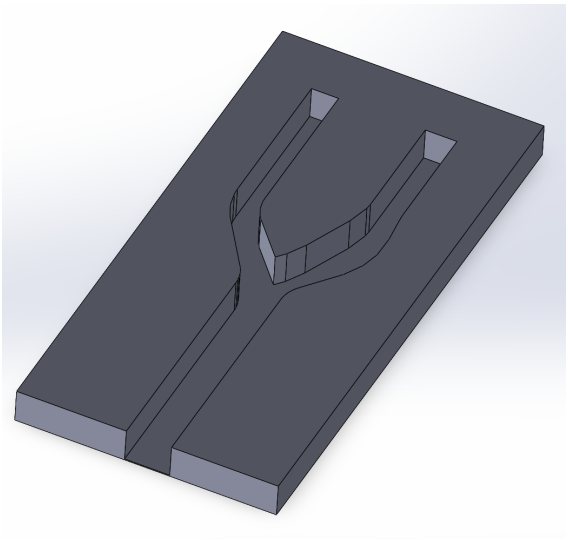
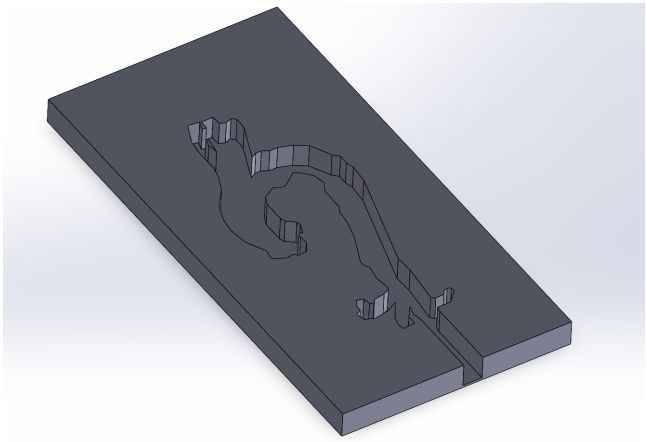


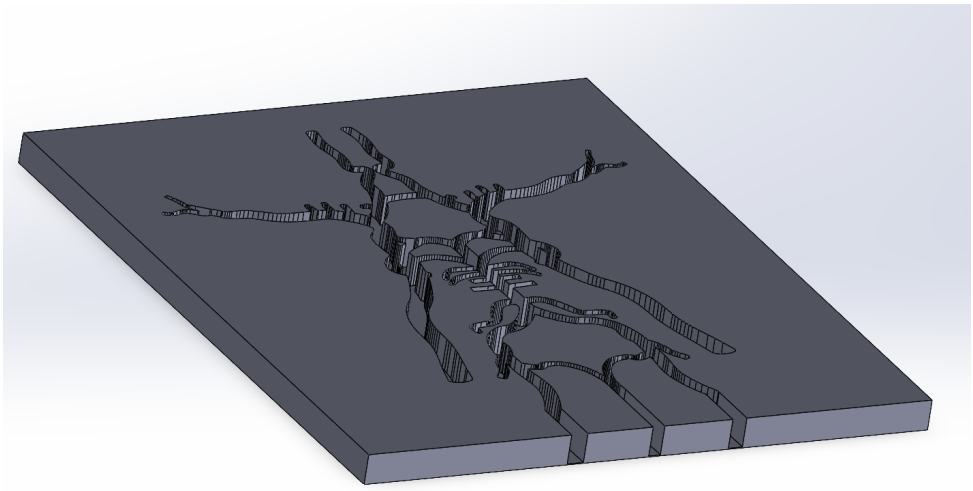
Figure A.7: Segmentation pipeline visualization.



(a)



(b)



(c)

Figure A.8: Additional phantom CAD models.

References

[1] N. Poojari et al. "StrokeNav2D: A Skeleton Based Dataset for Cerebrovascular Imitation Learning." *Design of Medical Devices Conference*, ASME, 2026, V001T07A003. doi:10.1115/DMD2026-1063.

[2] A. Berg, S. Khandagale, A. Imdieke, T. M. Kowalewski, A. Grande, S. L. Moen, and K. Desingh. "Vasculature Discretization and Search Space Simplification for Automating Catheter Navigation." *Design of Medical Devices Conference*, ASME, 2026, V001T02A007. doi:10.1115/DMD2026-1054.

[3] P.-H. Cheng et al. "Towards Remote Thrombectomy with Telerobotically-Driven Guidewires." *Design of Medical Devices Conference*, ASME, 2026, V001T07A005. doi:10.1115/DMD2026-1079.